John Ryan

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Summary

Senior Research Engineer with more than 8 years of machine learning experience, beginning with the development of production NLP systems in 2016. Combines a track record of founding deep-tech ventures with recent leadership in large-scale AI for drug discovery at Isomorphic Labs. Proven ability to lead technical projects from initial concept through to production.

Experience

ISOMORPHIC LABS | Senior Research Engineer - Performance and Scaling

Jan 2026 - Present | London, UK

| Research Engineer II - Applied ML Research

Nov 2024 - Dec 2025

| Research Engineer - Applied ML Research

Feb 2024 - Oct 2024

- **Executed core research** on frontier AI models, alongside developing advanced performance and scaling solutions to push the boundaries of computational efficiency.
- **Led** machine learning for two drug discovery programs, establishing a robust feedback loop between AI prediction and experimental validation to prioritize high-potential therapeutic candidates.
- **Architected and spearheaded** the development of a scalable internal platform for repeatable ML inference workflows for repeatable science, now used by 4+ teams.

ENTREPRENEUR FIRST | Founder in Residence

Oct 2023 - Jan 2024 | London, UK

- Selected for the globally competitive talent investor program to conceptualize and build a deep-tech venture from the ground up.

DUCTILE AI | Founder

July 2022 - Sep 2023 | London, UK

- **Led** the end-to-end design and development of a robust Visual SLAM system for resource-constrained aerial vehicles, from research to deployment.
- **Drove** the research strategy, evaluating and implementing novel approaches combining Neural Radiance Fields (NeRFs) and Bayesian Modelling.

Publications & Research

ISOMORPHIC LABS DRUG DESIGN ENGINE (ISODDE) | Technical Report | 2026

- Co-authored the technical report detailing the proprietary IsoDDE for end-to-end drug discovery, achieving state-of-the-art performance in protein-ligand structure prediction and binding affinity.

REINFORCEMENT LEARNING FOR SPACE OPERATIONS | MSc Thesis | 2022

Education

UNIVERSITY OF OXFORD Masters - Computer Science - Merit

Oct 2021 - June 2022

Bachelors - Computer Science - First Class Honours

Oct 2018- June 2021

Skills

Languages: Python, C++, CUDA, Golang, Javascript, Julia, Java, Scala, Bash
AI Frameworks & Libraries: PyTorch, JAX, Tensorflow, Triton, XLA, Pallas, Keras
Distributed Systems & MLOps: Docker, Kubernetes, DDP, FSDP, ROS, Linux, Spark